Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

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Dataset at a Glance

3,900

Purchases

18

Features

$59.76

Avg. Spend

3.75

Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season, Size

Behavior

Discounts, Frequency, Ratings, Shipping

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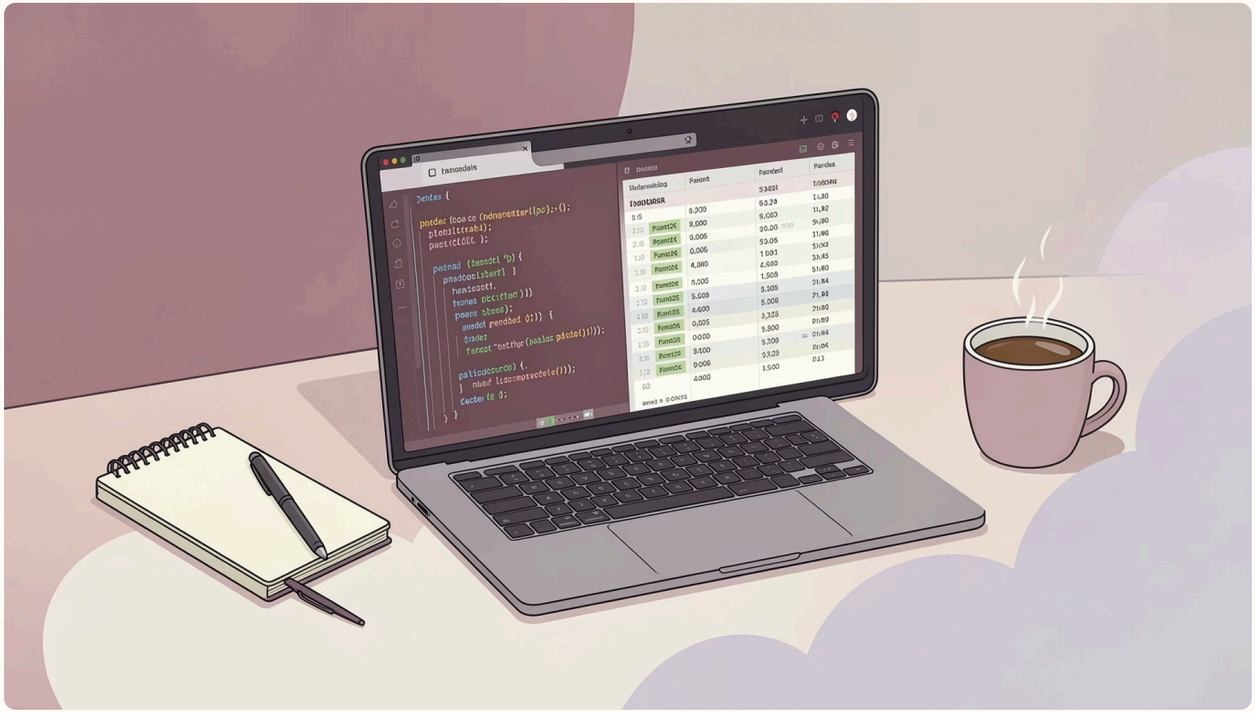
Data Preparation in Python



01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled 37 missing `Review Rating` values with category median

03

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Cleaned DataFrame exported to database for SQL analysis

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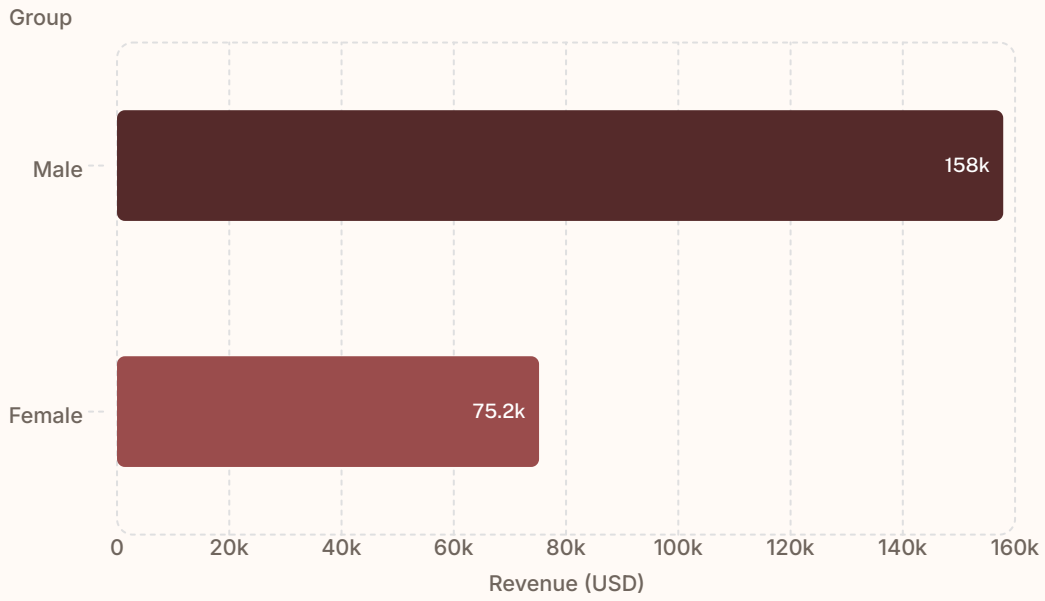
Revenue & Spending Insights



Key Findings

Male customers generate **2× more revenue** than female customers.

Express shipping users spend slightly more (**$60.48**) vs. Standard (**$58.46**).

839 discount users still spent **above average** — high-value deal seekers.

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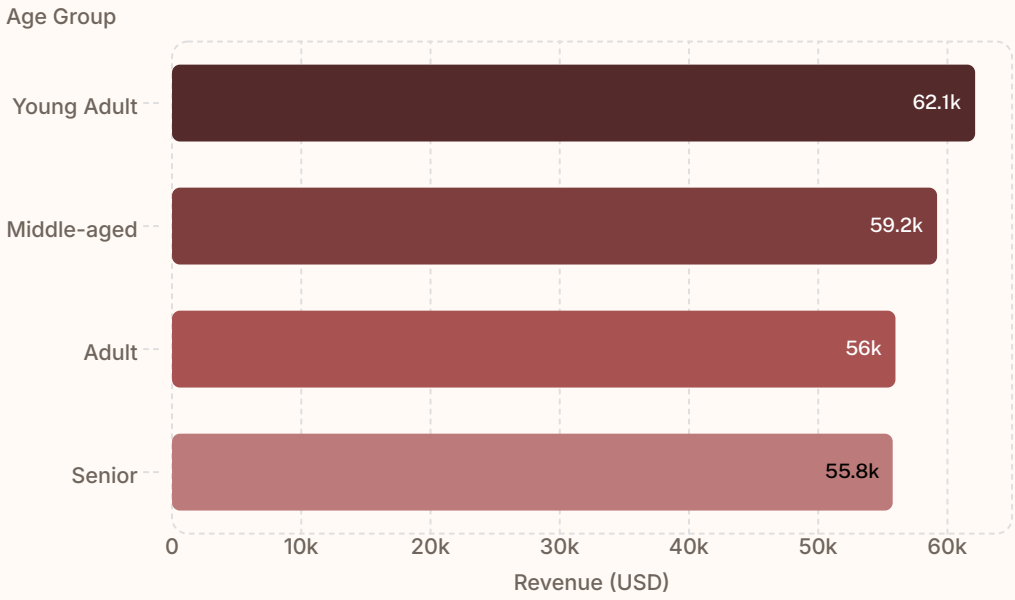
Revenue by Age Group



Young Adults Lead

Young Adults contribute the highest revenue at **$62,143**, followed closely by Middle-aged customers.

All age groups contribute within a **$6,400 range** — a relatively balanced distribution.

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Top Products & Categories

Accessories

Jewelry (171) · Sunglasses (161) · Belt (161)

Clothing

Blouse (171) · Pants (171) · Shirt (169)

Footwear

Sandals (160) · Shoes (150) · Sneakers (145)

Outerwear

Jacket (163) · Coat (161)

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Top-Rated Products & Discount Dependency

Highest Avg. Ratings

🥇 Gloves — 3.86

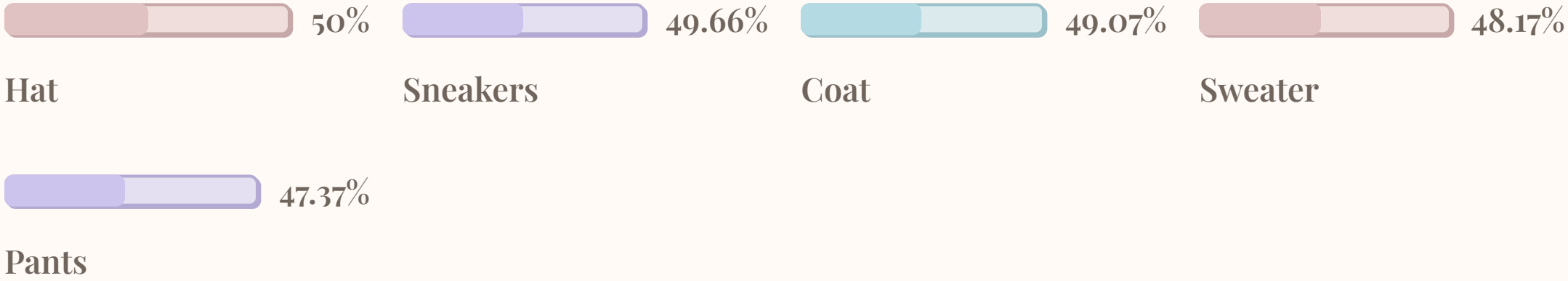
🥈 Sandals — 3.84

🥉 Boots — 3.82

Hat — 3.80

Skirt — 3.78

Most Discount-Dependent Items



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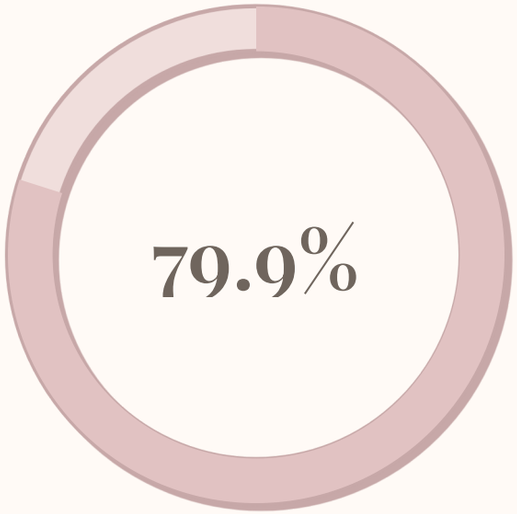
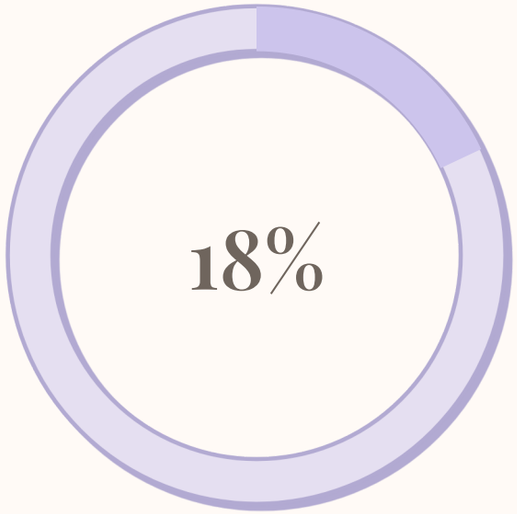
Customer Segmentation & Subscriptions



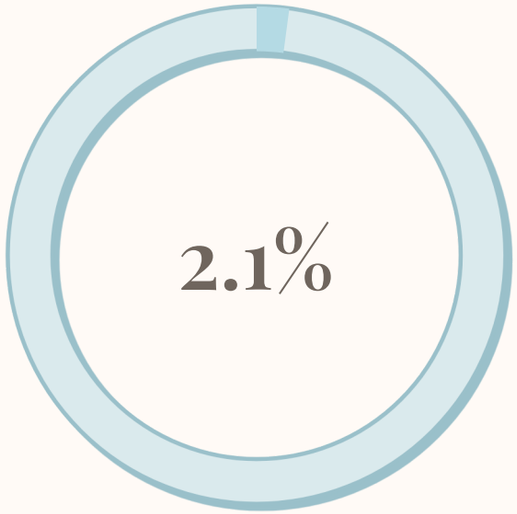
Loyal

3,116 customers



Returning

701 customers



New

83 customers

Subscribers vs. Non-Subscribers

Status	Customers	Avg. Spend	Revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 non-subscribed — loyalty drives subscription opportunity.

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Power BI Dashboard

Interactive Filters

- Subscription Status, Gender
- Category, Shipping Type

Visualizations

- Subscription donut: **27% Yes / 73% No**
- Revenue & Sales by Category
- Revenue & Sales by Age Group

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert non-subscribers

Loyalty Programs

Reward repeat buyers to grow the Loyal segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on Young Adults and express-shipping users

Product Positioning

Highlight top-rated and best-selling items in campaigns

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